...ELECTRONIC DEVICES EVEN IN 'OFF' POSITION IS TREATED AS EXAM MALPRACTICE

➢ DON'T WRITE ANYTHING ON THE QUESTION PAPER

Answer any <u>TEN</u> Questions

(10 X 10 = 100 Marks)

1. a) Write an assembly language code for the following expression using the IAS computer Instruction set. [6]

   A = (B+C) * D. Assume that the inputs are available in the memory location 450, 451 and 452. Store the results in memory location 750 and 751.

   b) Consider a hypothetical 32-bit microprocessor having 32-bit instructions composed of two fields: the first byte contains the opcode and the remainder of the immediate operand or an operand address. Discuss the impact on the system speed if the microprocessor bus has a 32-bit local address bus and a 16-bit local data bus. How many bits are needed for the program counter and the instruction register? [4]

2. a) Having multiplicand as 13 and multiplier as (-7), find the product using Booth's algorithm with appropriate flow diagram and step by step description. [7]

   b) Represent 1250.125_{10} as floating-point number in single precision format. [3]

3. a) Consider two different machines, with two different instruction sets, both of which have a clock rate of 200 MHz. The following measurements are recorded on the two machines running a given set of benchmark programs: [6]

Instruction Type	Instruction Count (millions)	Cycles per instructions
Machine A	8	1
	4	3
	2	4
	4	3
Machine B	10	1
	8	2
	2	4
	4	3

Determine the effective CPI, MIPS rate, and execution time for each machine. Comment on the results.

b) Enumerate and explain the procedure in the phases of Instruction cycle with a state diagram. **[4]**

4. a) A two-way set-associative cache has lines of 16 bytes and a total size of 8 Kbytes. The 64-Mbyte main memory is byte addressable. Show the format of main memory addresses. **[6]**

   b) Discuss the memory hierarchy in terms of size, access time and cost per bit. **[4]**

5. In cache memory management policies, explain the process of the page replacement algorithm FIFO with the cache lines of 4 frames and 6 frames. Show each step in detail and comment on the hit ratio and miss ratio obtained for both the frames.

   1 2 3 4 2 1 5 6 2 1 2 3 7 6 2 3 7 2 3 6

6. How many 1024x8 RAM chips are needed to provide a memory capacity of 2048x8? Draw the address map table and the chip organization with memory connection to CPU.

7. a) A computer system needs to transfer a large block of data from RAM to an external device such as hard drive. Explain how DMA can improve data transfer in this scenario. Also show the DMA configurations with relevant diagrams. **[7]**

   b) DMA controller has a transfer rate of 10 MB/s. If it takes 100 milliseconds to transfer a block of data, what is the size of the block? **[3]**

8. With a flow diagram, explain the interrupt driven Input Output Module. Discuss the different interrupt handling mechanisms and brief out the Interrupt overhead.

9. a) Discuss the salient features of the Magnetic Disk with its structural representation. List out the pros and cons of the magnetic disk. **[6]**

   b) Assuming an even parity, detect and correct the errors in the 7-bit hamming code 1011011. **[4]**

10. a) Differentiate block striping and mirroring Redundant Array of Independent Disk. [4]

b) A company's IT department wants to ensure data integrity and fault tolerance for its critical data storage. They have a large number of disks available for the array. Which RAID level would be the most appropriate choice, and what factors would you consider in making your recommendation? [6]

11. Classify Flynn's taxonomy of parallel computer architecture with suitable examples of each.

12. a) In a pipelined processor, a sequence of instructions is being executed. The third instruction requires the result of the first instruction as an operand. However, the result of the first instruction is not yet available when the third instruction needs it. Identify the type of hazard occurring in this scenario and propose a solution to mitigate it. [5]

b) Compare and contrast superscalar versus super pipeline architecture. [5]

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Faculty Name	**Prof. Sindhuja M**	Slot	**A2+TA2**
		Class Nbr	**CH2022232500336**
Time	**3 Hours**	Max. Marks	**100**

Part A (10 X 10 Marks)

Answer All questions

01. (a) Explain and illustrate any 3 bus structures available. **[10]**
(b) Compare and differentiate between Von Neumann and Harvard Architecture.

02. Explain 5-bit booth multiplier using the flowchart and find the result to multiply $(-9) \times (-13)$. **[10]**

03. Divide 11 (eleven) by 3 (three) using non-restoring method and explain the steps involved. **[10]**

04. (a)Consider a MIPS computer with a processor speed of 1GHZ and each ALU instruction takes 3 clock cycles, branch/jump instruction takes 2 clock cycles, each short word (SW) instruction takes 4 clock cycles and each long word(LW) instruction takes 5 clock cycles. Consider a bench mark program which takes 200 million ALU instructions, 55 million branching instructions, 25 million SW instructions and 20 million LW instructions. Find (i) CPI and (ii) CPU time. **[10]**
(b) A two-byte relative mode branch instruction is stored in memory location 1000. The branch is made to the location 87. What is the effective address?

05. (a)Consider a system in which bus cycles takes 500 ns. Transfer of bus control in either direction, from processor to I/O device or vice versa, takes 250 ns. One of the I/O devices has a data transfer rate of 50 KB/s and employs DMA. Data are transferred 1 byte at a time. Suppose we employ DMA in a burst mode. That is, the DMA interface gains bus mastership prior to the start of a block transfer and maintains control of the bus until the whole block is transferred. For how long would the device tie up the bus when transferring a block of 128 bytes? Repeat the calculation for cycle-stealing mode. **[10]**
(b)Explain the interrupt priority system using polling method.

06. (a)A two-way set associative cache memory uses block of four words. The cache can accommodate a total of 2048 words from main memory. The main memory size is 128K X 32.Formulate all pertained information required to construct the cache memory (Tag, Index, Data, blocks, words). **[10]**
(b) What is the size of the cache memory?

07. The access time of a cache memory is 100 ns and that of main memory is 1000 ns. It is estimated that 80% of the memory requests are for read and the remaining 20% is for write. The hit ratio for read access is 0.9. A write through procedure is used. **[10]**
(a)What is the average access time of the system considering only memory read cycle?
(b)What is the average access time of the system for both read and write cycles?
(c)What is the hit ratio taking into consideration the write cycles?

08. (a) A bit stream 10011101 is transmitted using the standard CRC method. The generator polynomial is x^3+1. What is the actual bit string transmitted? Suppose the third bit from the left is inverted during transmission. How will receiver detect this error ? [10]

(b) Explain the organization of magnetic disk with Read / Write Mechanism.

09. (a) What is Flynn's classification of computer architecture ? and explain each classification with block diagram. [10]

(b) Consider a pipeline having 4 phases with duration 60, 50, 90 and 80 ns. Given latch delay is 10 ns. Calculate

i) Pipeline cycle time

ii) Pipeline time for 1000 tasks

iii) Sequential time for 1000 tasks

10. (a) Explain the difference between superscalar and super pipeline architecture with block diagram. [10]

(b) Consider a non-pipelined processor with a clock speed of 2.5 MHz and average cycles per instruction of 4. The same processor is upgraded to a five stage pipelined RISC processor but due to the internal pipeline delay, the clock speed is reduced to 2 MHz Assume there are no stalls in the pipeline. Calculate

i) Cycle Time in Non-Pipelined Processor

ii) Non-Pipeline Execution Time

iii) Cycle Time in Pipelined Processor

iv) Pipeline Execution Time

v) Speed Up ratio

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Vellore Institute of Technology

1393/1395/1403/1417/1419/1421/1423/1425/1427/ 1431/1435/1441/1445/1449/1454/1476

Slot: C1+TC1

Time: Three Hours

Max. Marks: 100

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Answer ALL Questions

(10 X 10 = 100 Marks)

1. Perform an assembly language code and register transfer notation to evaluate the following arithmetic expression using IAS instructions.

$$S = (A - (B+C)) \times (D - (E+F))$$

Assume that data variables A, B, C, D, E, F are available at memory locations 200, 201, 202, 203, 204, 205 respectively. And S will be stored 500 onwards.

2. Illustrate Non- Restoring division algorithm and apply the same to divide the dividend 23 by the divisor 5. Show how the q_0 bit is replaced in each iteration.

3. i. Consider the expression $T = P + (Q - R) \times S$ and assume that data will occupy 24 bits, address occupy 24 bits and word length is 1 byte. **[6]**

   a) Evaluate the above expression for 3 address and 2 address machines.
   b) Calculate the memory to store and encode each of these sets in bytes.
   c) Calculate memory traffic in terms of memory accesses to fetch and execute the instruction.

   ii. Assume two different processors A and B executes the same instruction set with clock rate 5GHz and 6GHz respectively. The cycles per instruction of A is 2.0 and B is 3.5. Which processor among A and B performs best in terms of instructions per second? **[4]**

4. A computer employs RAM chips of 256 × 8 and ROM chips of 128 × 8. Design a computer system that needs 256 x 16 of RAM, 256 x 8 of ROM, and two interface units with 256 registers each. A memory mapped I/O configuration is used. The two higher -order bits of the address bus are assigned 00 for RAM, 01 for ROM, and 10 for interface registers.

   a) How many chips are needed in total for the design?
   b) Design a memory interface address map for the above system.
   c) Show the chip layout for the above design.

5. Explain how DMA transfers data to and from the peripherals with necessary sketch and Illustrate the different data transfer methods in DMA.

6. Delineate disk stripping? Illustrate the RAID levels that uses different types of stripping. Discuss their advantages and disadvantages.

7. i. Elaborate on Flynn's Taxonomy that involves multiple instructions. Highlight the role of shared and distributed memory in parallel machines. **[6]**

   ii. Differentiate super pipeline architecture from scalar pipeline architecture? **[4]**

8. Consider the following sequence of instruction in a 4-stage (F, D, E, W) pipelined execution. Sketch the space - time diagram. Identify the hazard that occurs while executing the instructions and explain the techniques to overcome this hazard.

MOV A, B	[A <-B]
ADD C, B, A	[C <- A + B]
SUB C, #5	[C<- C – 5]

9.a) i. Apply bit-pair recoding table for booth multiplication to find the recoded multiplier bits of 13 and use the recoded multiplier bits to multiply 31×13. **[5]**

   ii. Apply Booth Multiplication and show how it works efficiently to multiply 18 and -8. **[5]**

OR

9.b) i. With a neat sketch, explain the IEEE-754 standard floating point formats. Represent 6784.234_{10} in single precision and double precision format. **[6]**

   ii. Explain the steps involved in performing floating point multiplication and division. **[4]**

10.a) Consider a cache of 128 blocks of 256 words each and a byte addressable main memory consisting of 16,384 blocks.

   i. Calculate the number of bits required to address a main memory block.

   ii. Find the split of the physical address bits assuming the cache is direct, fully associative and 4-way set associative mapped.

   iii. Analyse the impact of using these mapping techniques for the above scenario.

OR

10.b) i. Consider the following sequence of memory references. 4, 0, 4, 2, 1, 6, 2, 5, 0, 5, 1, 7, 6, 2, 3. **[8]**

   Calculate hit ratio and miss ratio using FIFO and LRU with 4 cache lines.

   ii. Consider a computer system that employs a cache with an access time of 45 ns and a main memory with a cycle time of 550 ns. Suppose that the hit ratio for reads is 75%, what would be the average access time for reads if the cache is a "look-Aside" cache? **[2]**

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1. a) Explain about the Registers and Register file. [4]

   b) Write an assembly code for computing M=(N-P)*Q expression using IAS instructions [6]

   Assume that inputs are available in the memory locations 601 and 602,603. Store the result in memory locations 604 and 605.

2. a) Multiply the following pair of signed numbers -9*-13 using modified Booth Algorithm [6]

   b) Describe the IEEE representation of floating point numbers and represent 85.125 in single precision format. [4]

3. List out the Boolean expressions for control signals for an example instruction execution and investigate the components required for hardwired control logic. Design the Hardwired based control unit.

4. a) Our favourite program runs in 12 seconds on computer A, which has a 4GHz clock. We are trying to help a computer designer build a computer, Which will run this program in 6 seconds. The designer has determined that a substantial increase in the clock rate is possible, but this increase will affect the rest of the CPU design, causing computer B to require 1.5 times as many clock cycles as computer A for this program. What clock rate should we tell the designer to target? [5]

   b) Write the code to implement the expression A= (B-C)*D on 3 address and 2 address machines. Do not rearrange the expression. In accordance with programming language practice, computing expression should not change the values of its operands. And also compute the total memory traffic in bytes for both instruction fetch and instruction execution for code and implement the expression evaluation for the above said two machines. Assume that opcode occupy one byte, address occupies two bytes, data values occupy two bytes and word length one byte. [5]

5. A computer employs RAM chips of 128 x 8 and ROM chips of 512 x 8. The computer system needs 256 x 16 of RAM, 1024 x 16 of ROM, and two interface units with 256 registers each.

   a) Compute total number of decoders are needed for the above system?
   b) Design a memory-address map for the above system
   c) Show the chip layout for the above design with a neat sketch.

6. a) A block-set associative cache memory consists of 128 blocks divided into four block sets. The main memory consists of 16,384 blocks and each block contains 256 eight-bit words. [6]

i. How many bits are required for addressing the main memory?

ii. How many bits are needed to represent the TAG, SET and WORD fields?

b) Assume that a computer system employs a cache with an access time of 20ns and a main memory with a cycle time of 200ns. Suppose that the hit ratio for reads is 90%. [4]

i. What would be the average access time for reads if the cache is a "look-through" cache?

ii. What would be the average access time for reads if the cache is a "look-Aside" cache?

7. Draw the typical block diagram of a DMA controller and explain the various transfer modes that are used for direct data transfer between memory and peripherals.

8. Discuss various bus arbitrations methods followed to handle the priority request with the necessary diagrams.

9. a) Consider the input data to be transmitted is 1011010, Construct the encoded data using the hamming code method. [5]

b) Assume the data received is 10100010000 which contains an error, Identify the error bit position and correct the bit using hamming error correction and detection technique. [5]

10. Discuss in details, The RAID levels. Will RAID level increase the performance? Justify in details.

11. Explain data hazard and the ways to detect and handle the data hazards in the below mentioned sequences.

```
sub $2, $1,$3
and $12,$2,$5
or $13,$6,$2
add $14,$2,$2
sw $15,100($2)
```

12. a) Consider the execution of a program which results in the execution of 2 million instructions on a 500-MHz processor. The instruction mix and the CPI for each instruction type are given in below the table. [5]

Compute MIPS (Million Instructions Per Second) Rating using given data.

Instruction Type	CPI	Instruction Mix
Arithmetic and Logic	1	60%
Load/Stored with cache hit	2	18%
Branch	4	12%
[illegible]	[illegible]	10%